AI-Lingo: System Requirements Specification (Refined)

Draft. Items marked Proposed or "to be confirmed" are not final until the team approves them.

1. Introduction

1.1 Purpose

This document specifies the functional requirements, non-functional requirements, interfaces, and constraints of AI-Lingo. It is the basis for design, implementation, and testing. Each requirement has a priority and an acceptance criterion so it can be verified.

1.2 Scope

AI-Lingo is a Flutter mobile app that helps learners improve their English through AI-generated, personalized conversational scenarios. The initial release provides text-based dialogue practice, vocabulary review, progress tracking, and offline access to saved content. Other languages and advanced features are future work.

1.3 Objectives

- O1 Personalize Language Learning: generate conversational scenarios based on learners' interests, goals, and practical needs.
- O2 Adapt Dialogue Complexity: use available learner information and, where implemented, assessment responses to guide dialogues toward an appropriate proficiency level.
- O3 Apply Prompt Engineering: use Prompt Chaining and Constrained Prompting to structure AI generation, prioritize previously learned vocabulary, and introduce relevant new expressions.
- O4 Support Real-World Practice: provide instructional dialogues based on everyday, academic, and professional situations.
- O5 Track Learning Progress: preserve learned vocabulary, progress information, and selected lesson content for continued access.

- O6 Ensure Maintainability and Data Protection: apply modular object-oriented design, structured state management, local persistence, and appropriate security mechanisms.

1.4 Definitions

Term	Meaning
CEFR	Common European Framework of Reference levels (A1 to C2)
Learned word	A saved word the learner has marked as known (exact rule to be confirmed)
Scenario	A short real-world situation (e.g., ordering food, a job interview) that frames a dialogue
Lesson snippet	A saved excerpt of a generated dialogue

1.5 Priority Key

M = Must (required for release), S = Should (expected), C = Could (optional), Proposed = extra item pending team approval.

2. Functional Requirements

2.1 Learner Profile

ID	Requirement	Pri	Acceptance criterion
FR-01	The system shall let the learner set and edit interests, learning goals, and a proficiency level.	M	The profile can be created and edited, and is saved after restarting the app.
FR-02	The system shall optionally estimate the learner's level from assessment responses. The learner can override the result.	C	If implemented, the learner sees a suggested level and can change it.

2.2 Content Generation

ID	Requirement	Pri	Acceptance criterion
FR-03	The system shall generate practical scenarios (everyday, academic, professional) based on the learner's interests, goals, and level.	M	Generated scenarios relate to the learner's interests or goals.
FR-04	The system shall generate an instructional dialogue for the selected scenario at the learner's level.	M	The dialogue difficulty matches the selected level.
FR-05	The system shall use previously learned vocabulary in dialogues and introduce a limited number of relevant new expressions.	M	Dialogues reuse learned words and highlight the new expressions.
FR-06	The system shall check the format of AI responses before showing them.	Proposed	Badly formatted responses are not shown to the learner.
FR-07	The system shall apply content-safety checks to generated content.	Proposed	Inappropriate content is not shown to the learner.

FR-17 (Must): Prompt Engineering. The system shall use Prompt Chaining and Constrained Prompting to generate scenarios and dialogues. How they are applied is to be reviewed with the team.

2.3 Conversational Practice

ID	Requirement	Pri	Acceptance criterion
FR-08	The system shall provide a text-based chat interface for practicing with AI-generated dialogues.	M	The learner can reply and receive the next AI turn on the same screen.
FR-09	The system shall give brief corrections on learner replies.	Proposed	Replies with errors receive a short correction.

ID	Requirement	Pri	Acceptance criterion
FR-10	The learner shall be able to end a session, and the session shall be recorded in progress data.	M	Ended sessions appear in the progress history.

2.4 Vocabulary, Progress, and Saved Content

ID	Requirement	Pri	Acceptance criterion
FR-11	The learner shall be able to save new words or expressions from a dialogue.	M	Saved words appear in the vocabulary list after restarting the app.
FR-12	The learner shall be able to review saved vocabulary and mark words as learned.	M	The learned status is updated after review.
FR-13	The system shall store and show learning progress (sessions completed, words saved, words learned).	M	The progress screen matches the stored data.
FR-14	The learner shall be able to save lesson snippets, reopen them, and delete them.	M	A saved snippet can be reopened and deleted.

2.5 Offline Behavior and Errors

ID	Requirement	Pri	Acceptance criterion
FR-15	Without internet, the learner shall be able to access saved vocabulary, progress data, and saved lesson snippets.	M	With airplane mode on, these features work normally.
FR-16	When generation is unavailable (offline or service error), the system shall show a clear message and let the learner retry later.	M	The message appears and no saved data is lost.

3. Non-Functional Requirements

ID	Category	Requirement	Pri	Verification
NFR-01	Maintainability	The code shall use a modular object-oriented design, with interfaces and mixins where appropriate.	M	Code review.
NFR-02	State management	The app shall use Cubit/BLoC for application state and learning progress.	M	Code review.
NFR-03	Security	Locally stored learning data shall be encrypted, and the encryption key shall be protected with the Android Keystore. The encryption method is to be reviewed with the team.	M	Stored data is not readable outside the app.
NFR-04	Privacy	Only the data needed for generation shall be sent to the AI provider.	Proposed	Review of the AI requests.
NFR-05	Performance	Local screens shall open quickly, and a loading indicator shall show while content is generated.	Proposed	Manual testing.
NFR-06	Reliability	Saved data shall not be lost when the app restarts or updates.	M	Restart and update tests.
NFR-07	Usability	A new learner shall be able to set up a profile and start a dialogue without help.	Proposed	Informal user test.

ID	Category	Requirement	Pri	Verification
NFR-08	Compatibility	The minimum Android version is to be decided.	Proposed	Test on the chosen versions.

4. External Interfaces

- **4.1 User interface:** text-based screens for profile, scenario selection, chat practice, vocabulary, progress, and saved snippets.
- **4.2 AI service interface:** accessed via an abstract interface so the provider can be changed. Provider and API are **to be decided**. The interface shall support timeout, structured output, and error codes.
- **4.3 Local storage interface:** Hive NoSQL boxes for vocabulary, progress, snippets, and profile.
- **4.4 Security interface:** Android Keystore for the encryption key (see NFR-03).

5. Technical Constraints

1. Flutter and Dart.
2. Object-oriented design with interfaces and mixins where appropriate.
3. Cubit/BLoC for state management.
4. Hive NoSQL for local persistence.
5. Encryption and Android Keystore as in NFR-03.
6. Initial release: English only, text only.
7. Content generation requires network access and the selected AI service.

6. Out of Scope (Initial Release)

1. Real-time voice conversations.
2. Pronunciation scoring.
3. Fully offline AI generation.
4. Languages other than English.
5. Advanced adaptive learning beyond the personalization described here.

7. Open Decisions

#	Decision	Needed for
1	AI provider and API	4.2, FR-03 to FR-05
2	Whether the proficiency assessment is in the initial release	FR-02
3	Encryption method	NFR-03
4	How Prompt Chaining and Constrained Prompting are applied	FR-17
5	Minimum Android version, and whether iOS is planned	NFR-08, 4.4
6	User accounts, backup, and sync	Section 6
7	Rule for a "learned" word	FR-12, FR-13
8	Whether to include the Proposed items (FR-06, FR-07, FR-09, NFR-04, NFR-05, NFR-07)	Section 10

Items marked Proposed are not confirmed until the team approves them.

8. Requirements Traceability

Objective	Requirements
O1 Personalization	FR-01, FR-03
O2 Adaptive complexity	FR-02, FR-04
O3 Prompt Engineering	FR-05, FR-17
O4 Real-world practice	FR-03, FR-04, FR-08
O5 Progress tracking	FR-11 to FR-15, NFR-06
O6 Maintainability and data protection	NFR-01 to NFR-04

9. Summary of Changes from v1.0

- Added priorities and simple acceptance criteria.
- Kept the core requirements from the Project Overview.
- Removed numeric targets until the team confirms them.
- Restored Prompt Engineering as FR-17.
- Marked extra items as Proposed (see Section 10).

10. Proposed Additions (Not in the Project Overview)

These items go beyond the Project Overview and are marked in the text. Keep, change, or remove each one.

- FR-06: format check of AI responses.
- FR-07: content-safety checks.
- FR-09: corrections on learner replies.
- NFR-04, NFR-05, NFR-07: privacy, performance, and usability.
- NFR-08: minimum Android version.
- Offline access to progress data (the Overview names only saved vocabulary and lesson snippets).
- User accounts, backup, and sync: not in the Overview, so left as open decision #6.